

Architecture Comparison

XBandDigitalModule_20x20 - Architecture Comparison

Document Information

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1. Executive Summary

This document compares four RF sampling architectures for the XBandDigitalModule_20x20, evaluating trade-offs in performance, complexity, power, and risk. The Direct RF Sampling architecture is se

2. Architecture Options

2.1 Option 1: Direct RF Sampling (SELECTED)

Description: ADC samples X-band signal directly at 10.4 GSPS.

RF In (8-12 GHz) -> BPF -> LNA -> Balun -> ADC (10.4 GSPS) -> FPGA
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Aspect	Evaluation
Components	Minimal (BPF, L
**ADC Requireme	10.4 GSPS, 8 GH
**Clock Require	10.4 GHz, <100
**FPGA Requirem	16-lane JESD204
SNR	~55 dB (limited
Flexibility	Highest (softwa
Size	Smallest
Power	72-96 W
Complexity	Low (analog), H
Risk	Medium (new tec
Heritage	Growing (TI TID

Pros:

Cons:

2.2 Option 2: Bandpass Sampling

Description: ADC samples at sub-Nyquist rate, using aliasing intentionally.

RF In (8-12 GHz) -> BPF -> LNA -> ADC (2-4 GSPS) -> FPGA
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Aspect	Evaluation
Components	Minimal (BPF, L
**ADC Requireme	2-4 GSPS, 8 GHz
**Clock Require	2-4 GHz, <200 f
**FPGA Requirem	8-lane JESD204C
SNR	~45-50 dB (degr
Flexibility	Low (fixed band
Size	Small
Power	40-60 W

Complexity	Medium (aliasin
Risk	Medium-High
Heritage	Limited for X-b

Pros:

Cons:

2.3 Option 3: Analog Downconversion

Description: Mixer translates X-band to IF, then ADC samples IF.

RF In (8-12 GHz) -> BPF -> LNA -> Mixer -> IF BPF -> ADC (1-2 GSPS) -> FPGA
?
LO (8-12 GHz synthesizer)

Aspect	Evaluation
Components	Many (Mixer, LO
**ADC Requireme	1-2 GSPS, 500 M
**Clock Require	1-2 GHz, <500 f
**FPGA Requirem	4-lane JESD204C
SNR	~65-70 dB (best
Flexibility	Low (fixed LO f
Size	Medium
Power	20-40 W
Complexity	High (analog),
Risk	Low (traditiona
Heritage	Excellent (deca

Pros:

Cons:

2.4 Option 4: Hybrid IF Digital

Description: Two-stage conversion with digital IF processing.

RF In (8-12 GHz) -> BPF -> LNA -> Mixer -> IF (500 MHz) -> ADC (1 GSPS) -> FPGA	
?	?
LO (first)	Clock

Aspect	Evaluation
Components	Moderate (Mixer
**ADC Requireme	1 GSPS, 500 MHz
**Clock Require	1 GHz, <500 fs
**FPGA Requirem	4-lane JESD204C
SNR	~60-65 dB
Flexibility	Medium
Size	Medium
Power	25-45 W
Complexity	Medium (analog
Risk	Low-Medium
Heritage	Good

Pros:

Cons:

3. Comparison Matrix

Criterion	Weight	Direct RF	Bandpass	Downconv	Hybrid	Winner
SNR Performance	20%	7	5	9	7	Downconv
Flexibility	15%	10	4	3	5	Direct RF
Component Count	15%	9	8	4	6	Direct RF
Size	10%	9	8	5	6	Direct RF
Power	10%	5	7	9	7	Downconv

Risk	15%	6	4	9	7	Downconv
Heritage	15%	6	3	9	7	Downconv
Weighted Scor	**100%	**7.35**	**5.55**	**6.85**	**6.35**	**Direct RF**

4. Component Availability

4.1 ADC Options

Architecture	Required ADC	Available	Status
Direct RF	10+ GSPS, 8 GHz	ADC12DJ5200-SP	? Available
Bandpass	2-4 GSPS, 8 GHz	ADC12DJ3200QML-	? Available
Downconv	1-2 GSPS, 500 M	EV12AQ600 (6.4	? Available
Hybrid	1 GSPS, 500 MHz	ADC12DJ3200QML-	? Available

4.2 FPGA Options

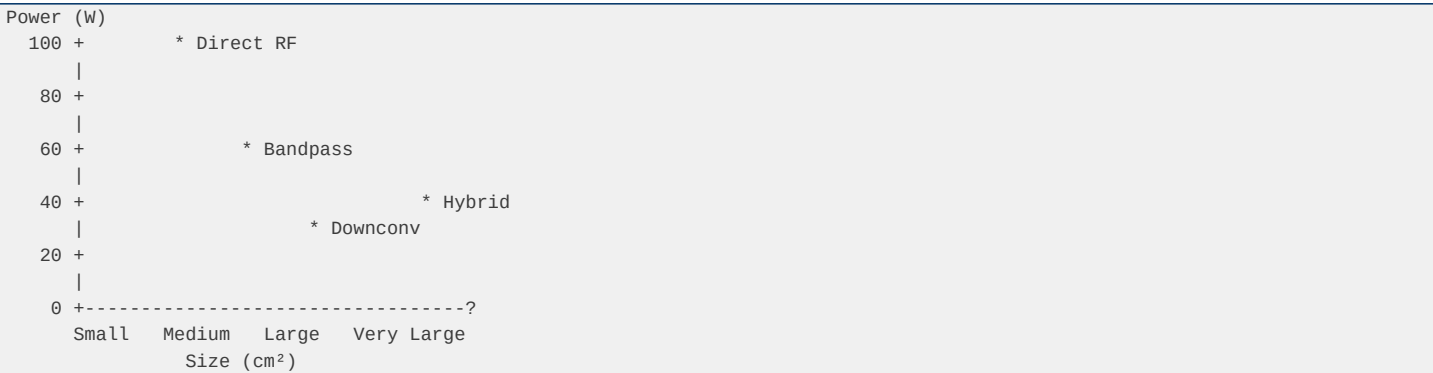
Architecture	Required FPGA	Available	Status
Direct RF	16+ Gbps transc	XQRVC1902 (26.5	? Available
Bandpass	10+ Gbps transc	XQRKU060 (12.5	? Available
Downconv	5+ Gbps transce	XQRKU060	? Available
Hybrid	5+ Gbps transce	XQRKU060	? Available

5. Trade-off Analysis

5.1 SNR vs. Flexibility



5.2 Size vs. Power



5.3 Risk vs. Heritage

Heritage (years)	
30 +	* Downconv



6. Sensitivity Analysis

6.1 If SNR is Most Critical

Winner: Analog Downconversion (Option 3)

6.2 If Size is Most Critical

Winner: Direct RF Sampling (Option 1)

6.3 If Risk is Most Critical

Winner: Analog Downconversion (Option 3)

7. Recommendation

7.1 Selected Architecture: Direct RF Sampling

Rationale:

7.2 Risk Mitigation

To address the higher risk of direct RF sampling:

7.3 Alternative Recommendation

If direct RF sampling risk is unacceptable:

8. Revision History

Version	Date	Author	Description
1.0	2026-09-09	System Engineer	Initial release
